) 2022 Contingency.pdf

<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>award (1) for any of following
extraction (of aluminium) is very expensive
extraction is much more expensive than recycling
recycling is much cheaper than extraction

award (1) for reasoning
because much larger amounts of electricity/energy needed

award (1) for reference to conserving the ore if no other mark awarded</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td></td><td><table><tr><th>Left-hand side of the membrane</th><th>Right-hand side of the membrane</th><th></th></tr><tr><td>Na^+ Cl^- H^+ OH^-</td><td>Cl^- H^+ OH^-</td><td></td></tr><tr><td>H^+ OH^- Cl^-</td><td>Na^+ H^+ OH^-</td><td></td></tr><tr><td>Na^+ Cl^- H^+ OH^-</td><td>Na^+ H^+ OH^-</td><td>✓</td></tr><tr><td>Na^+ Cl^-</td><td>Na^+ OH^-</td><td></td></tr></table></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td><table><tr><td>titanium burns in chlorine in aqueous conditions</td><td></td></tr><tr><td>titanium doesn't react with chlorine under any conditions</td><td></td></tr><tr><td>aqueous conditions prevent chlorine from reacting with titanium</td><td>✓</td></tr><tr><td>aqueous conditions prevent chlorine from reacting with sodium chloride</td><td></td></tr></table></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr>
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